

Lab program-40:

40. Illustrate the various File Access Permission and different types of users in Linux.

Code:

```
#include <stdio.h>
#include <sys/stat.h>
#include <sys/types.h>
#include <unistd.h>

int main(void) {
    const char *filename = "file.txt";
    FILE *file = fopen(filename, "w");
    if (file == NULL) {
        perror("fopen");
        return 1;
    }
    fputs("Linux file permission example.\n", file);
    fclose(file);

    mode_t permissions = S_IRUSR | S_IWUSR | S_IRGRP | S_IWGRP | S_IROTH;
    if (chmod(filename, permissions) == -1) {
        perror("chmod");
        return 1;
    }

    printf("File permissions changed successfully.\n");
    printf("Use 'ls -l %s' to view owner, group and permission details.\n", filename);
    return 0;
}
```

OUTPUT:

1. Compile the C program (assuming it's saved in a file named `change_permissions.c`):

bash

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```
gcc -o change_permissions change_permissions.c
```

1. Run the program:

bash

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```
./change_permissions
```

Output:

If the program executes successfully, it should display the following output:

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```
File permissions changed successfully.
```